# World Financial Cycles and Global Trade


Yan Bai
U of Rochester, CUHK, NBER, and CEPR

Minjie Deng
Simon Fraser U

Chang Liu
Stony Brook U

Gabriel Mihalache
Ohio State


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# The Question

- Emerging market spreads rise and fall together: the Global Financial Cycle
- Usual suspects are financial: US monetary policy, global risk appetite
- Miranda-Agrippino and Rey (2022): the second global factor tracks global trade
- **Can global trade shocks generate a world financial cycle?**

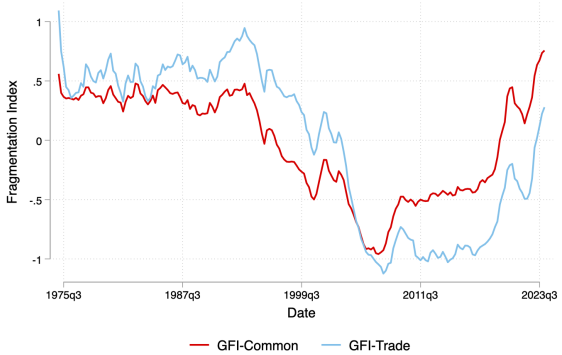
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- Two facts to explain: when global trade cost rises, spreads
  - *rise* on average and *become more dispersed* across countries

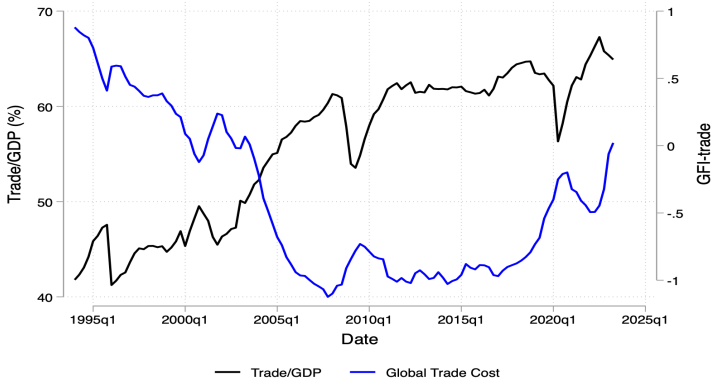
- **Common:** costlier imported inputs everywhere, spreads rise together
  - Under balanced trade, this is the whole story: equal losses, no change in dispersion
- **Differential:** *high debt countries save*, deeper recession, depreciate more
  - High debtors' spreads rise by more, dispersion widens

# Geopolitical Fragmentation Index




- Trade cost proxied with quarterly *Geopolitical Fragmentation Trade Index* (GFI-trade)
  - Fernandez-Villaverde et al. (2024): dynamic hierarchical factor model
  - 88 countries data $\rightarrow$ 16 global variables, 4 categories

# Global Trade and Trade Cost

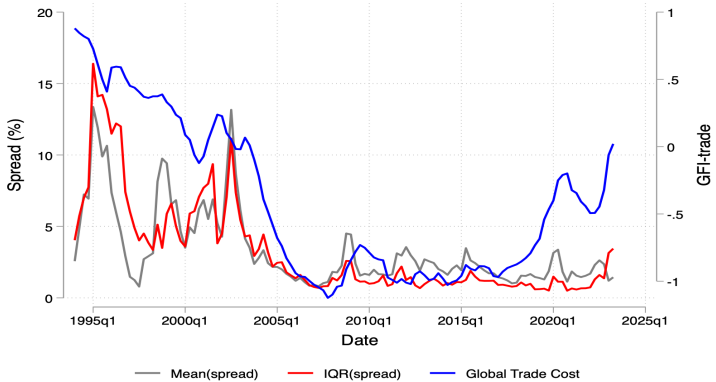



Falling global trade cost accompanied by larger trade flows

Global trade cost measured by GFI-Trade index from Fernández-Villaverde, Mineyama and Song (2025)

35 countries including small OECD + emerging countries

# Global Trade Cost and Spread




Falling global trade cost accompanied by lower & less dispersed sovereign spreads

*How do global trade shocks affect countries differentially?*

# This Paper

- Global trade barriers in emerging markets with unbalanced trade and default risk
- **Quantitatively:** *world GE model*, one advanced economy, many emerging markets
  - Gross trade and sovereign default with endogenous borrowing terms
- **Empirically:** *heterogeneous effect panel local projection*, 36 countries
  - Higher global trade cost: high debt countries lose more output, spreads rise more

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- **General equilibrium is central:** a global shock hits all emerging markets together
  - They cut imports and try to save at once: global demand and the world interest rate move
  - This loops back to the advanced economy, then to emerging markets via borrowing costs
  - *A world financial cycle driven by trade: spreads rise together and fan out*
- **Other experiments and counterfactuals:**
  - Advanced economies' tariffs, export subsidies, transitory vs permanent shocks, ...

# A Model of Global Trade and Financial Cycle

**World general equilibrium model:** global trade + sovereign default

- Rep. advanced economy (AE) and a continuum of emerging markets (SOE)
- Gross trade flows: all import final goods from all, use as intermediates
- Sovereign default setting for SOEs: incomplete markets, default
- Global + idiosyncratic iceberg trade costs

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**Mechanism**: Following a *global trade cost shock* $\uparrow$

- Intermediate goods $\downarrow \Rightarrow$ output $\downarrow \Rightarrow$ sovereign default risk $\uparrow$ (**same across countries**)

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**Mechanism**: Following a *global trade cost shock* ↑

- Intermediate goods ↓ ⇒ output ↓ ⇒ sovereign default risk ↑ (**same across countries**)
- *High-debt countries* need to save: imports fall more than export
  ⇒ Greater output drop, greater depreciation
  ⇒ Disincentivize debt repayment
  ⇒ Sovereign default risk ↑ more (**heterogeneous across countries**)

# Preview of Results

- Parameterize using cross-sectional/business cycle financial & trade moments
- Model results consistent with data:
  - Panel local projection with heterogeneous effects
  - Global trade costs $\uparrow \Rightarrow$ high-debt countries: greater output $\downarrow$, greater spread $\uparrow$
- Counterfactual: Stand-in advanced economy imposes tariffs on SOEs
  $\Rightarrow$ AE larger deficit, higher SOE spreads, larger global imbalances

- **Sovereign default.** Eaton and Gersovitz (1981); Aguiar and Gopinath (2006); Arellano (2008); Cuadra et al.(2010); Yue (2010); Hatchondo et al.(2016); Chatterjee and Eyigungor (2012, 2015); Na et al. (2018), Aguiar et al. (2016), Aguiar and Amador (2023), etc.
  Introduce trade frictions as a driver of dynamics in sovereign spreads

- **Global financial cycles:** Longstaff et al. (2011); Miranda-Agrippino and Rey (2020, 2022); Morelli et al. (2022); Gilchrist et al. (2022); Bai et al. (2024), etc.
  Aggregate trade shocks impact trend and time-varying dispersion of spreads

- **World G.E. models with int'l financial markets**: Bai and Zhang (2012); Morelli et al. (2022); Alessandria et al. (2024), etc.
  World GE model of interactions between gross trade and limited enforcement

- **Trade costs consequences of sov defaults:** Rose (2005); Borensztein and Panizza (2010); Martinez and Sandleris (2011); Mendoza and Yue (2012); Zymek (2012); Serfaty (2021); Andreasen et al. (2024); Alessandria et al. (2024), etc.
  Impact of sovereign defaults on trade, through global equilibrium effects

# A Model of Gross Trade Flows and Default

# Model

- World general equilibrium model:
  - Measure $\xi$ continuum of small open economies (SOE), discount factor $\beta$
  - One stand-in advanced economy (AE), discount factor $\beta_{AE} > \beta$
- Countries
  - Produce using labor $L$, domestic $H$ and imported intermediate good $M$

    $$Q_{it} = z_{it} L_{it}^{\alpha} \left( \omega_s H_{it}^{\frac{\gamma-1}{\gamma}} + (1-\omega_s) M_{it}^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma\theta}{\gamma-1}}$$

  - Iceberg trade costs $\tau_t \epsilon_{it} p_t^M M_{it}$
- Imperfect international financial markets
  - Centralized borrowing and default decision
  - Sovereign default $\Rightarrow$ haircut + productivity loss (no market exclusion)
- Idiosyncratic productivity shock $z_{it}$, trade cost shock $\epsilon_{it}$ + global trade cost shock $\tau_t$

# Individual Country (SOE): Sovereign

- Country $i$ issues claims to $b_{it+1}$ units of long-term defaultable bonds at $t$
  - Sequence of payments starting from $t+1$: $\kappa$, $\kappa(1-\delta)$, $\kappa(1-\delta)^2$,...
  - $\delta$: controls bond duration

- The sovereign decides whether to default and future bond position $b_{it+1}$
  - Default ($d_{it}=1$) $\Rightarrow$ *immediate* debt reduction, $b_{it} \to \phi b_{it}$, with $\phi < 1$
  - Cost: utility cost ($\psi$) + productivity loss ($\tilde{z}_{it} = h(z_{it}, d_{it}) \leq z_{it}$)

- Budget constraint

$$T_{it} = -\kappa \phi^{d_{it}} b_{it} + q_t(z_{it}, b_{it+1}) \left[ b_{it+1} - (1-\delta)\phi^{d_{it}} b_{it} \right]$$

# Advanced Economy

- Identical production structure as SOEs
- No idiosyncratic shocks
- No default, bond economy
- Later, AE import tariffs (+ export tax or subsidy in paper, other experiments)

# Global Intermediate Producer

- Competitive global intermediaries assemble goods from all countries:

$$Y_t = \left( X_{\text{AE},t}^{\frac{\eta-1}{\eta}} + \xi \int_0^1 X_{it}^{\frac{\eta-1}{\eta}} di \right)^{\frac{\eta}{\eta-1}}$$

- Optimization

$$\max_{X_{\text{AE},t},\{X_{it}\}} p_t^M Y_t - \xi \tau_t \int_0^1 p_{it} \epsilon_{it} X_{it} di - \underbrace{p_{\text{AE},t}}_{=1} \tau_t X_{\text{AE},t}$$

$\Rightarrow$ demand function and price index for global intermediate

$$X_{AE,t} = \left( \frac{\tau_t}{p_t^M} \right)^{-\frac{1}{\eta}} Y_t, \qquad X_{it} = \left( \frac{\tau_t \epsilon_{it} p_{it}}{p_t^M} \right)^{-\frac{1}{\eta}} Y_t, \qquad p_t^M = \tau_t \left( 1 + \xi \int_0^1 (\epsilon_{it} p_{it})^{1-\eta} di \right)^{\frac{1}{1-\eta}}$$

# Static Private Trade Equilibrium

Given the trade cost $\tau_t$ and distribution $\Omega_t(\tilde{z}_{it}, \epsilon_{it}, T_{it})$, the static-private equilibrium allocations $\{C_{it}, L_{it}, Q_{it}, X_{it}, M_{it}\}$, prices $\{w_{it}, p_{it}, p_t^M\}$ satisfy the following

- Firms' optimization, for each $i$ and AE
- Households optimizes, for each $i$
- Global intermediate producers optimize
- Goods market clearing and balance of payment, for each $i$

$$Q_{it} = C_{it} + H_{it} + \tau_t \epsilon_{it} X_{it}, \qquad T_{it} + \underbrace{p_{it} \tau_t \epsilon_{it} X_{it}}_{\text{Exports}} = \underbrace{p_t^M \tau_t \epsilon_{it} M_{it}}_{\text{Imports}}$$

- Global intermediate goods clearing, $Y_t = \tau_t \left( M_{\text{AE},t} + \xi \int_0^1 \epsilon_{it} M_{it} di \right)$

# Dynamic World Equilibrium: AE

- AE: trades assets internationally, full commitment
- Euler equation

$$u_{\mathrm{AE},t} q_t^{\mathrm{rf}} = \beta_{\mathrm{AE}} u_{\mathrm{AE},t+1} \left( \kappa + (1-\delta) q_{t+1}^{\mathrm{rf}} \right)$$

- Define short-term risk-free rate

$$R_t = \frac{1}{\beta_{\mathrm{AE}}} \frac{u_{\mathrm{AE},t}}{u_{\mathrm{AE},t+1}}$$

# Dynamic World Equilibrium: SOE

Given aggregate state variable $S_t = (\tau_t, \Omega_t(z_{it}, \epsilon_{it}, b_{it}))$, government $i$ solves

- Default decision, at the start of the period,

$$V_{it}(z_{it}, \epsilon_{it}, b_{it}, S_t) = \max_{d_{it} \in \{0,1\}} \{d_{it} \left[W_{it}(\tilde{z}_{it}, \epsilon_{it}, \phi b_{it}, S_t) - \psi\right] + (1 - d_{it}) W_{it}(z_{it}, \epsilon_{it}, b_{it}, S_t)\}$$

- Then, new asset position

$$\begin{aligned} W_{it}(z_{it}, \epsilon_{it}, b_{it}, S_t) = \max_{b_{it+1}} & \{u(C_{it}, L_{it}) + \beta \mathbb{E}_t V(z_{it+1}, \epsilon_{it+1}, b_{it+1}, S_{t+1})\} \\ \text{s.t. } & T_{it} = -\kappa b_{it} + q_t(z_{it}, b_{it+1}) \left[b_{it+1} - (1 - \delta) b_{it}\right], \\ & C_{it} = \mathcal{C}_{it}(z_{it}, \epsilon_{it}, T_{it}, \Omega_t, \tau_t), \quad L_{it} = \mathcal{L}_{it}(z_{it}, \epsilon_{it}, T_{it}, \Omega_t, \tau_t), \\ & \Omega_{t+1} = H_\Omega(\Omega_t, \tau_t) \end{aligned}$$

# Bond Pricing

- Intl' lenders: risk neutral and competitive, fully committed
  - $b_{it+1} \leq 0$: Country $i$ holds long-term bonds issued by international lenders
  - $b_{it+1} > 0$: Lenders hold long-term risky bonds issued by country $i$
- Introduce short-term, gross risk-free rate $R_t$
- Equilibrium bond prices:

$$q_{it} = \begin{cases} q_t^{rf}, & \text{if } b_{it+1} \leq 0 \\ \frac{1}{R_t} \mathbb{E}_t \left[ \phi^{d_{it+1}} \left( \kappa + (1-\delta) q_{it+1} \right) \right], & \text{otherwise} \end{cases}$$

with $q_t^{rf} = \frac{1}{R_t} \left( \kappa + (1-\delta) q_{t+1}^{rf} \right)$

# Market Clearing

- Goods market clearing and balance of payment, for each $i$

$$Q_{it} = C_{it} + H_{it} + \tau_t \epsilon_{it} X_{it},$$

$$\underbrace{p_{it}\tau_t\epsilon_{it}X_{it}}_{\text{Exports}} - \underbrace{p_t^M\tau_t\epsilon_{it}M_{it}}_{\text{Imports}} = \underbrace{\kappa\phi^{d_{it}}b_{it} - q_t(z_{it}, b_{it+1})\left[b_{it+1} - (1-\delta)\phi^{d_{it}}b_{it}\right]}_{\text{govt transfer } (-T_{it})}$$

- Global intermediate goods clearing,

$$Y_t = \tau_t \left( M_{\text{AE},t} + \xi \int_0^1 \epsilon_{it} M_{it} di \right)$$

- Bonds market clearing:

$$q_t^{\text{rf}} b_{\text{AE},t} + \xi \int q_t(z_{it}, \epsilon_{it}, b_{it+1}) b_{it+1} d\Omega_t = 0.$$

# Differential Effects of Global Trade Shocks

- Intuition from static trade equilibrium, for fixed trade balances ($T_{it}$)
- The Keynes–Ohlin "transfer problem"
- $T_{it} < 0$: countries who need to save (usually high-debt countries)

# Static Private Equilibrium

Given $\{\tau, Y, p^M\}$ and $(z, T)$ for each country, each SOE's $\{C, L, H, M, X, Q, p\}$ solve

$$X = \left(\frac{\tau p}{p^M}\right)^{-\eta} Y \qquad \text{(export demand)}$$

$$\alpha Q/L = \chi L^{1/\nu} \qquad \text{(labor market)}$$

$$\tau p^M/p = \theta(1-\omega) Q G^{\frac{1}{\gamma}-1} M^{-\frac{1}{\gamma}} \qquad (M \text{ demand})$$

$$1 = \theta\omega Q G^{\frac{1}{\gamma}-1} H^{-\frac{1}{\gamma}} \qquad (H \text{ demand})$$

$$Q = z L^{\alpha} G(H, M)^{\theta} \qquad \text{(gross output)}$$

$$C = Q - H - \tau X \qquad \text{(domestic resources)}$$

$$T + \tau p X = \tau p^M M \qquad \text{(BoP)}$$

Similarly for AE conditions, but $p_{AE} = 1$ and drop BoP
Global intermediate goods clearing $Y = \tau \left(M_{\text{AE}} + \xi \int M(z, T) d\Omega(z, T)\right)$

# Mechanism: Effects of Trade Cost Shocks

Following a trade cost shock $\tau \uparrow$,

- Output $Q$, exports, imports $\downarrow$ for all countries $(z, T)$

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Following a trade cost shock $\tau \uparrow$,

- Output $Q$, exports, imports $\downarrow$ for all countries $(z, T)$
- Under balanced trade, $T = 0$, and all countries (SOEs & AE) have same $\%\triangle Q$
  - Independent of country size

# Mechanism: Effects of Trade Cost Shocks

Following a trade cost shock $\tau \uparrow$,

- Output $Q$, exports, imports $\downarrow$ for all countries $(z, T)$
- Under differential trade balances,
  Greater saving incentives ($T < 0$) $\Rightarrow$ $Q \downarrow$ more, $p \downarrow$ (depreciate) more

$$\underbrace{-T/p}_{\text{saver}>0} = \tau X - \tau p^M M \approx \tau X - (1-\theta)Q$$

  - Imports (and thus $Q$) $\downarrow\downarrow$, exports $X = \left(\tau p / p^M\right)^{-\eta} Y$ fall by less (thus $p \downarrow\downarrow$)
  - Fall in $p$ further increases real trade surplus $\Rightarrow$ *worse recession* $Q$

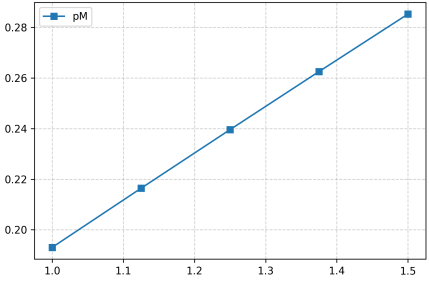
  - **Disincentivize gov't from repaying debt**

# Simple Numerical Illustration

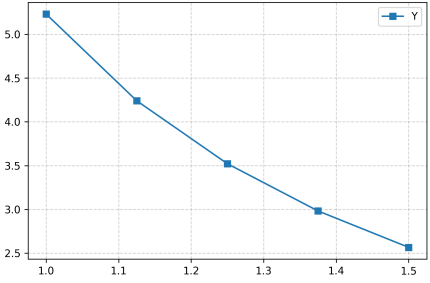
- Fix standard parameter values:
  $\alpha = 0.4, \theta = 0.43, \chi = 1.0, \nu = 0.72, \eta = 3, \gamma = 1, \omega = 0.6$

- Equilibria with $\tau \in [1, 1.5]$
  - Total transfer equals zero, $T_{AE} + \zeta \sum T_i = 0$
  - $T_i < 0$: net exporter (*larger saving incentives*)

# Global Trade Shock: Average Effect



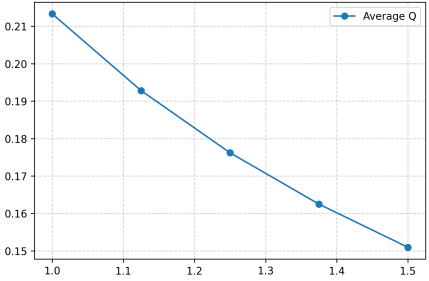

$\tau \Rightarrow p^M$



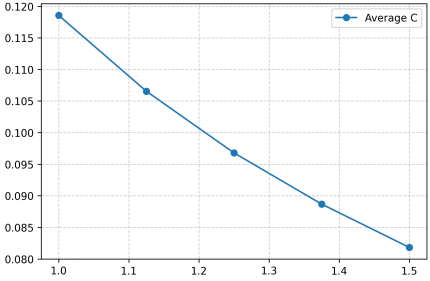

$\tau \Rightarrow Y$

- Higher trade cost $\Rightarrow$ increase in global intermediate price $p^M$, fall in global demand $Y$

# Global Trade Shock: Average Effect



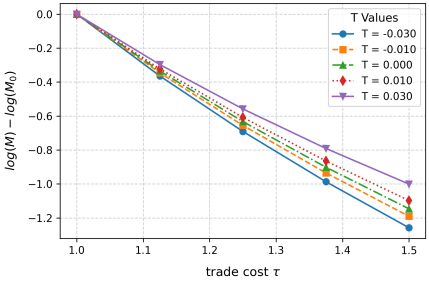

$\tau \Rightarrow Q$



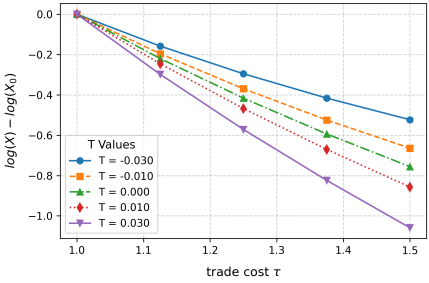

$\tau \Rightarrow C$

- Higher trade cost $\Rightarrow$ fall in output & consumption

# Global Trade Shock: Differential Effect



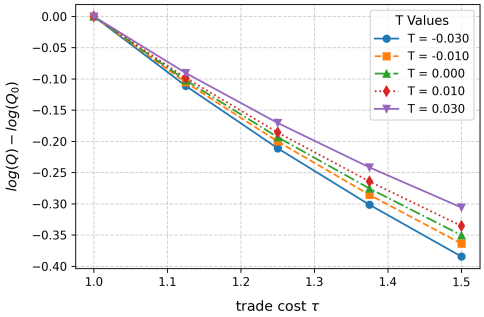

$\tau \Rightarrow M$



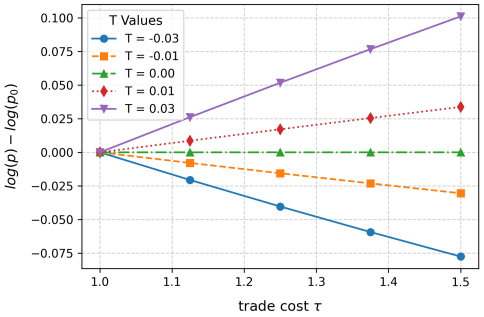

$\tau \Rightarrow X$

After $\tau \uparrow$, with grater savings $(T < 0) \Rightarrow$ *Imports falls by more, exports falls by less*

# Global Trade Shock: Differential Effect




$\tau \Rightarrow Q$

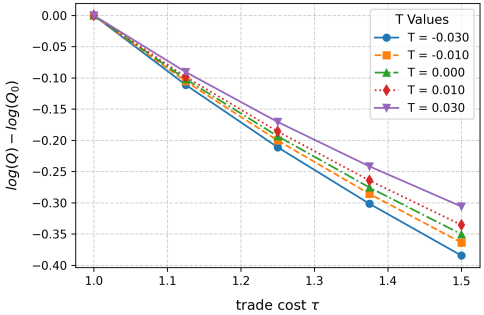



$\tau \Rightarrow p$

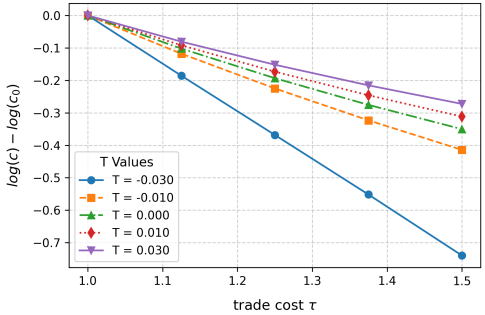
After $\tau \uparrow$, savings more $(T < 0) \Rightarrow$

- $Q$ falls by more, $p$ falls by more (greater depreciation)

# Global Trade Shock: Differential Effect




$\tau \Rightarrow Q$




$\tau \Rightarrow C$

After $\tau \uparrow$, savings more $(T < 0) \Rightarrow$

- $C$ falls by even more: $C = \theta Q + T/p$

# Full Model, Dynamic General Equilibrium

- Gov't internalizes the impact of $\tau$
- Endogenous borrowing and default decision
- Global bonds markets clear

# Quantitative Analysis

- Parameterize using trade and sovereign spreads moments
  - At the invariant distribution, $R = 1/\beta_{AE}$
  - Data for AE: consolidate USA + Germany + Japan + China
  - Data SOE: 36 countries, OECD small + emerging markets

- Differential effect after global trade cost shock
  - Which country exhibits greater spreads increases?

- One-time "MIT" transitory shock experiments
  - Glocal trade cost $\tau$, AE tariff, global v idiosyncratic trade costs, default v bond SOEs
  - More in the paper: transitory v permanent shocks, AE export tax or subsidy, ...

# Parameterization

| *Assigned Parameters* | |
|---|---|
| Risk aversion | $\sigma = 2$ |
| Elas. of substitution: | $\gamma = 3$ |
| Trade elasticity | $\eta = 3$ |
| Labor share | $\alpha = 0.40$ |
| Intermediate share | $\theta = 0.43$ |
| Frisch elasticity | $\nu = 0.72$ |
| Discount factor AE | $\beta = 0.995$ |
| Debt duration | $\delta = 1/16$ |
| Debt recovery rate | $\phi = 0.5$ |
| Productivity persistence | $\rho_z = 0.9$ |
| SOE trade cost persistence | $\rho_\epsilon = 0.9$ |

| *Parameters from Moment Matching* | |
|---|---|
| *Using moments from stationary dist.* | |
| Disutility of working | $\chi = 441.5$ |
| Import share of AE | $\omega_{AE} = 0.31$ |
| Import share of SOE | $\omega = 0.16$ |
| Mass of SOE | $\xi = 56.7$ |
| AE mean productivity | $\bar{z}_{AE} = 32.8$ |
| SOE mean productivity | $\bar{z}_{SOE} = 2.73$ |
| *Using moments in simulation* | |
| Discount factor SOE | $\beta = 0.959$ |
| Productivity volatility | $\sigma_z = 0.87\%$ |
| SOE trade cost volatlity | $\sigma_\epsilon = 2.2\%$ |
| Default costs | $\lambda_0 = -0.9$ |
| | $\lambda_1 = 0.945$ |

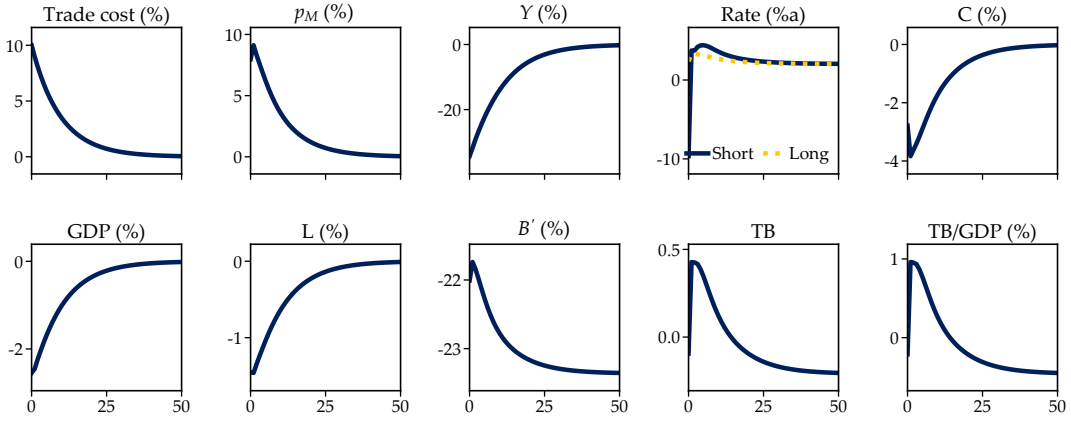
# Targeted Moments

| | Data | Model |
|---|---|---|
| *Moments from Stationary Dist. (%)* | | |
| AE share of world GDP | 49.5 | 49.5 |
| Average SOE GDP over $\text{GDP}_{AE}$ | 1.8 | 1.8 |
| Import share of GDP, AE | 16.2 | 16.2 |
| Import share of GDP, SOE | 29.4 | 29.4 |
| *Business cycle Moments, SOE (%)* | | |
| Std. real GDP | 5.2 | 6.0 |
| Mean spread | 1.0 | 2.1 |
| Mean Debt to output | 25.8 | 24.7 |
| Std. spread | 1.0 | 0.8 |
| Std consumption/std GDP | 1.0 | 1.0 |
| Corr(spread, import/GDP) | $-35$ | $-38$ |
| Corr(export/gdp, import/GDP) | 83 | 64 |

| | Value | Internal | Source or Target |
|---|---|---|---|
| Production and Trade | | | |
| $\zeta$ | 56.7 | × | AE share of World GDP |
| $z_{AE}$ | 32.8 | | Normalization of $Q_{AE}$ |
| $z_{SOE}$ | 2.73 | × | Relative GDP size of average SOE to AE |
| $\omega_{AE}$ | 0.31 | × | Import share AE |
| $\omega$ | 0.16 | × | Import share SOE |
| $\alpha$ | 0.4 | | Returns to scale, Labor |
| $\theta$ | 0.43 | | Return to scale, Intermediates |
| $\eta$ | 3.0 | | Imported intermediates substitution |
| $\gamma$ | 3.0 | | Domestic-imported intermediates substitution |
| $\rho_z$ | 0.9 | | Arellano et al. 2025 |
| $\rho_\epsilon$ | 0.9 | × | Corr spread, imports-to-GDP |
| $\sigma_z$ | $8.7e^{-3}$ | × | SOE GDP volatility |
| $\sigma_\epsilon$ | $2.2e^{-2}$ | × | Corr imports, exports |
| Preferences | | | |
| $\sigma$ | 2.0 | | Reference value |
| $\beta_{AE}$ | 0.995 | | Short-term risk-free rate |
| $\beta$ | 0.959 | × | Std log consumption to std log GDP ratio |
| $\chi$ | 441.5 | | Normalization of AE labor supply |
| $\nu$ | 0.72 | | Frisch elasticity, Healthcote et al. 2010 |
| Debt and Default | | | |
| $\delta$ | $16^{-1}$ | | Macaulay duration |
| $\lambda_0$ | −0.9 | × | Spread |
| $\lambda_1$ | 0.945 | × | Spread volatility |
| $\phi$ | 0.5 | | Haircut, excl. preemptive, Cruces and Trebesch 2013 |
| $\psi$ | 0.31 | × | SOE debt-to-GDP |
| $\bar{q}$ | 0.6 | | Underwriting stds., Chatterjee and Eiygongur 2012 |

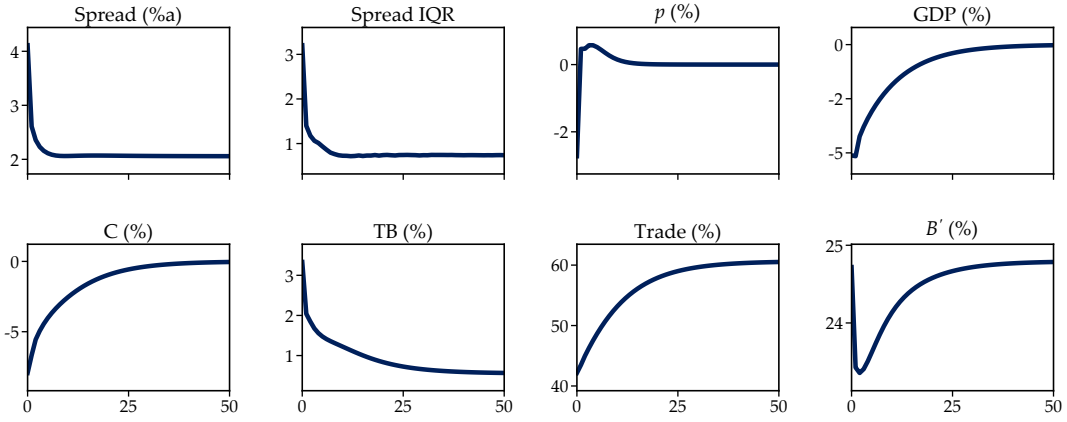
| | Data | Model | Targeted |
|---|---|---|---|
| Means | | | |
| AE Share of World GDP | 49.5 | 49.5 | × |
| Average SOE to AE GDP | 1.8 | 1.8 | × |
| Spread | 1.0 | 2.1 | × |
| Debt-to-GDP | 25.8 | 24.7 | × |
| Imports-to-GDP AE | 16.2 | 16.2 | × |
| Imports-to-GDP SOE | 29.4 | 30.0 | × |
| Trade Balance-to-GDP | 1.0 | 0.5 | |
| Standard Deviations | | | |
| Log GDP | 5.2 | 6.0 | × |
| Spread | 1.0 | 0.8 | × |
| Std Log C / Std Log GDP | 1.0 | 1.0 | × |
| Trade Balance-to-GDP | 3.5 | 2.3 | |
| Exports-to-GDP | 6.1 | 2.7 | |
| Imports-to-GDP | 6.0 | 2.8 | |
| Correlations | | | |
| Spread, Log GDP | −11 | −45 | |
| Spread, Trade Balance-to-GDP | 19 | 59 | |
| Spread, Exports-to-GDP | −12 | 11 | |
| Spread, Imports-to-GDP | −35 | −38 | × |
| Trade Balance-to-GDP, Log GDP | −18 | 24 | |
| Exports-to-GDP, Log GDP | 7 | 24 | |
| Imports-to-GDP, Log GDP | 19 | 3 | |
| Exports, Imports | 83 | 64 | × |

# Global Trade Cost Shock: AE



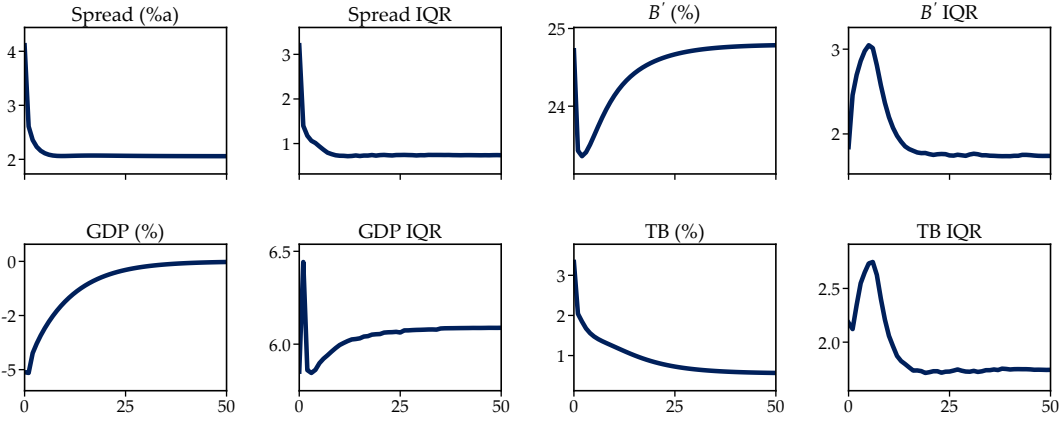

- Higher global trade shock $\tau \Rightarrow$ lower global intermediate production $Y$, raises price $p^M$
  lower AE GDP and consumption, higher interest rate

# Global Trade Cost Shock: SOE



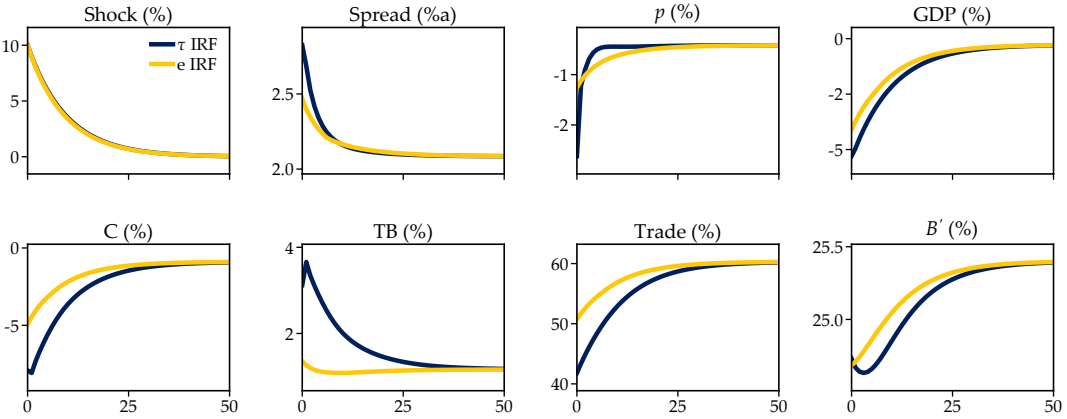

- Higher global trade shock $\tau \Rightarrow$ higher sovereign spreads, depreciation, lower GDP, trade surplus (TB$>$ 0) for SOE

# Global Trade Cost Shock: SOE Dispersions



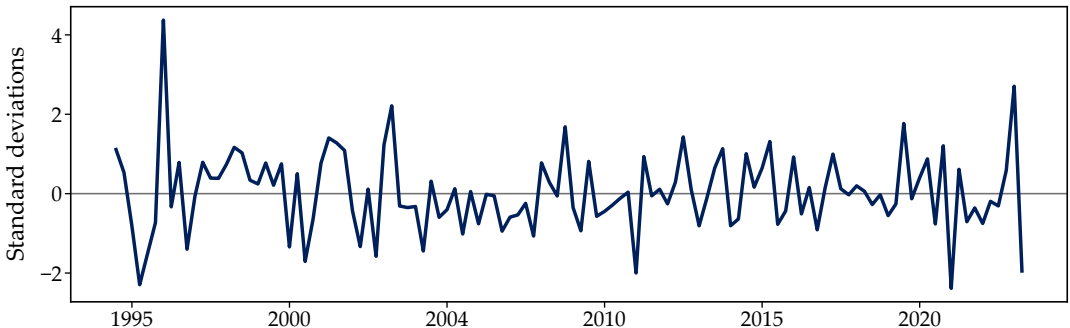

- Higher global trade shock $\tau \Rightarrow$ larger dispersion in spreads and trade balances among SOEs

# Global $\tau$ versus Idiosyncratic $e$ Trade Costs




- Same shock magnitude and persistence, both enter symmetrically in all equations.
  **Difference:** $\tau$ features *G.E. effects* via $p^M$, $Y$, and $R$

# Trade-Fragmentation Shock




The regressor $s_t$ is this SVAR innovation to GFI-trade, standardized to one standard deviation.

# Testing the Differential Effect of $\tau$

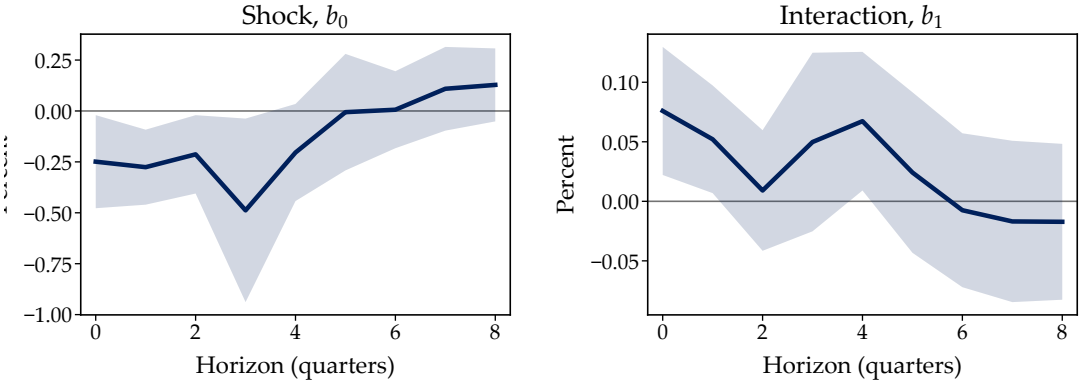
Unbalanced quarterly panel, 36 countries. At horizon $h = 0, \ldots, H$,

$$y_{i,t+h} = \alpha_i^h + b_0^h s_t + b_1^h (s_t \times z_{i,t-1}) + g^h z_{i,t-1} + \sum_{k=1}^{3} \phi_k^h \tilde{y}_{i,t-k} + u_{i,t+h}$$

Where

- $s_t$ is the GFI-Trade SVAR innovation. A positive value is greater trade fragmentation.
- $z_{i,t-1}$ is NIIP/GDP, demeaned within quarter & normalized by its pooled s.d.
- $b_0^h$ is the response at the mean position.
  $b_1^h$ is the additional effect, per one-s.d. higher NIIP/GDP.
- $y$ is HP-filtered log GDP or the log sovereign spread. $\tilde{y}$ is HP-filtered log GDP.

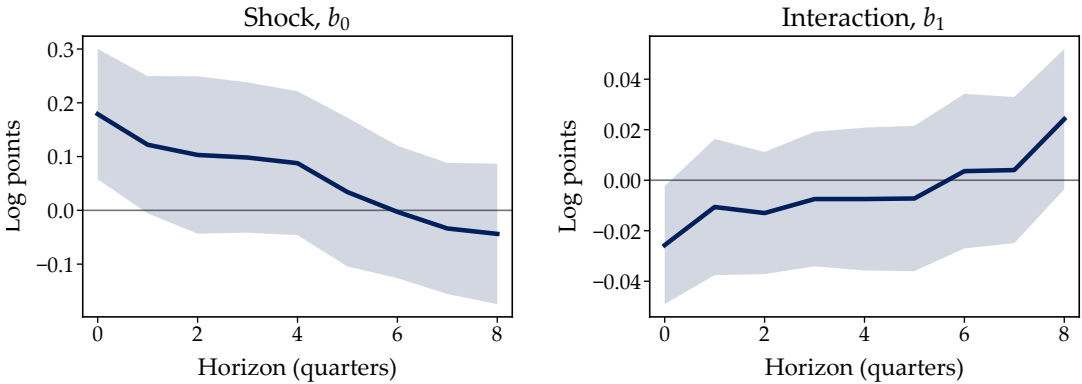
# Response of Real GDP




Percent deviation from an HP trend. Bands are 90% confidence intervals.

- Lower NIIP (more indebted) $\Rightarrow$ deeper recession.

# Response of Sovereign Spread




Log points. Bands are 90% confidence intervals.

- Lower NIIP (more indebted) ⇒ greater spread increase.

# Towards an Analysis of AE Tariffs

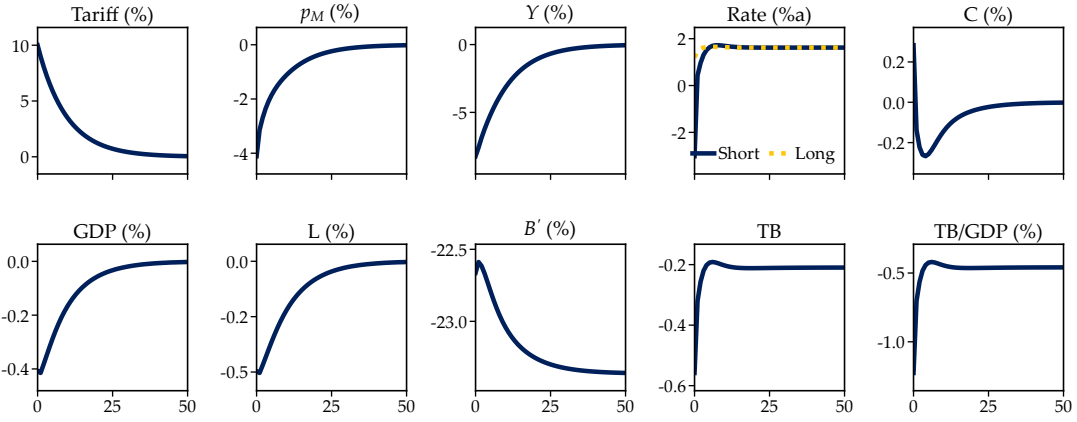
- Advanced economy imposes tariffs $\varepsilon^m_{AE}$
- Equilibrium conditions

$$\tau p^M (1 + \varepsilon^m_{AE}) = \theta(1-\omega) Q G^{\frac{1}{\gamma}-1} M^{-\frac{1}{\gamma}} \qquad (M \text{ demand})$$

Other conditions same as before, including

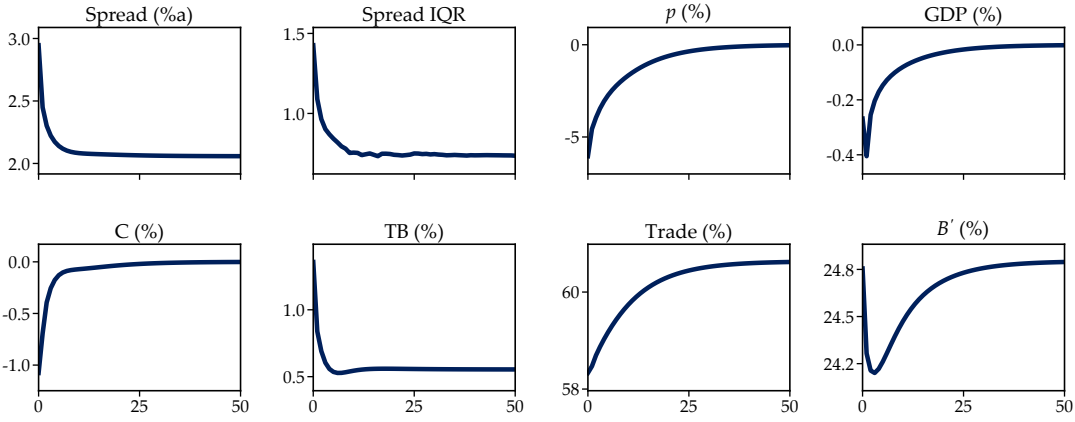
$$C_{AE} = Q_{AE} - H_{AE} - \tau X_{AE} \qquad \text{(domestic resources)}$$

# AE Tariff: AE



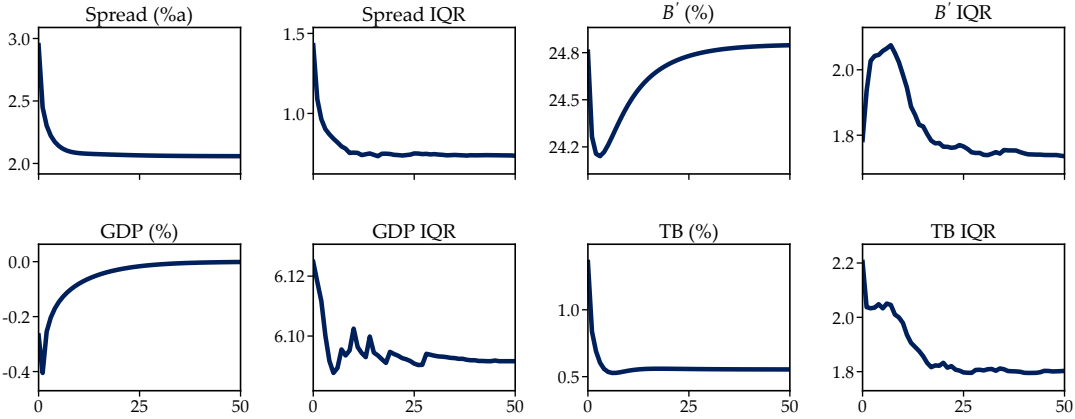

- Higher AE tariff $\Rightarrow$ lower global intermediate production $Y$, *lowers price* $p^M$
  *raise* AE consumption, *lower* interest rate

# AE Tariff: SOE



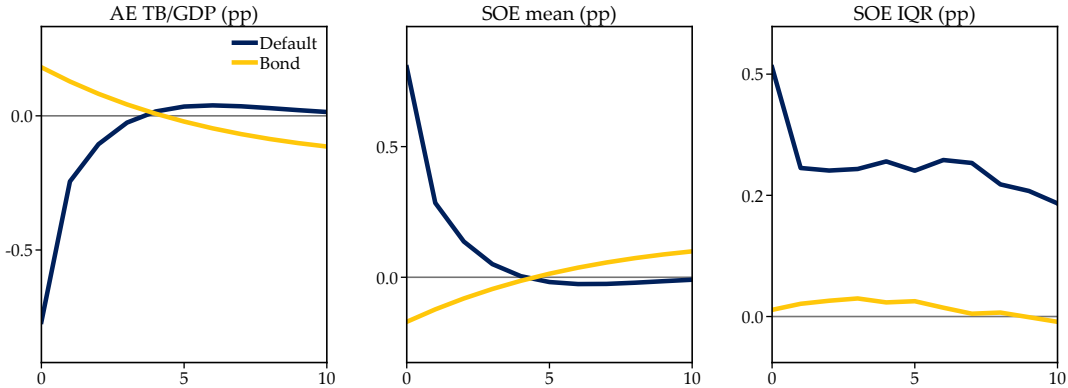

- Higher AE tariff $\Rightarrow$ higher sovereign spreads, depreciation,
  lower GDP, trade surplus (TB$>$ 0) for SOE
  Trade deficit for AE

# AE Tariff: SOE Dispersions



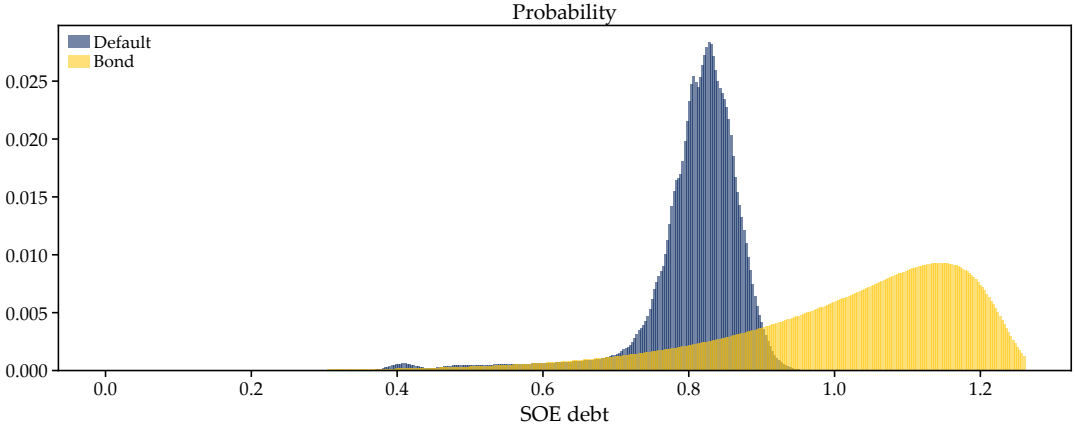

- Higher AE tariff $\Rightarrow$ larger dispersion in spreads, GDP, trade balances among SOEs

# AE Tariff: Default versus Risk-free Bond Economies




- **Can tariffs increase the trade balance of the AE?** SOEs' *financial friction severity* is key. Tight borrowing schedule from default risk? Then no.

# Default versus Risk-free Bond Economies: Debt Positions




- Risk-free bond economy with loose borrowing limit, plus patient SOEs with mass away from limit. Tariff hits, SOEs borrow and AE willing to save. With default, SOEs must export.

# Conclusion

- We build a world GE model with gross trade and sovereign default
- We find differential effects of global trade cost shocks
  - High-debt countries (savers): greater fall in output & larger increase in spreads
  - Explains movements in average and cross-country dispersion of spreads
- A framework to study impact of tariffs from advanced economies under imperfect international financial markets

# Appendix

# Default and Bond Economy: Trade Cost

